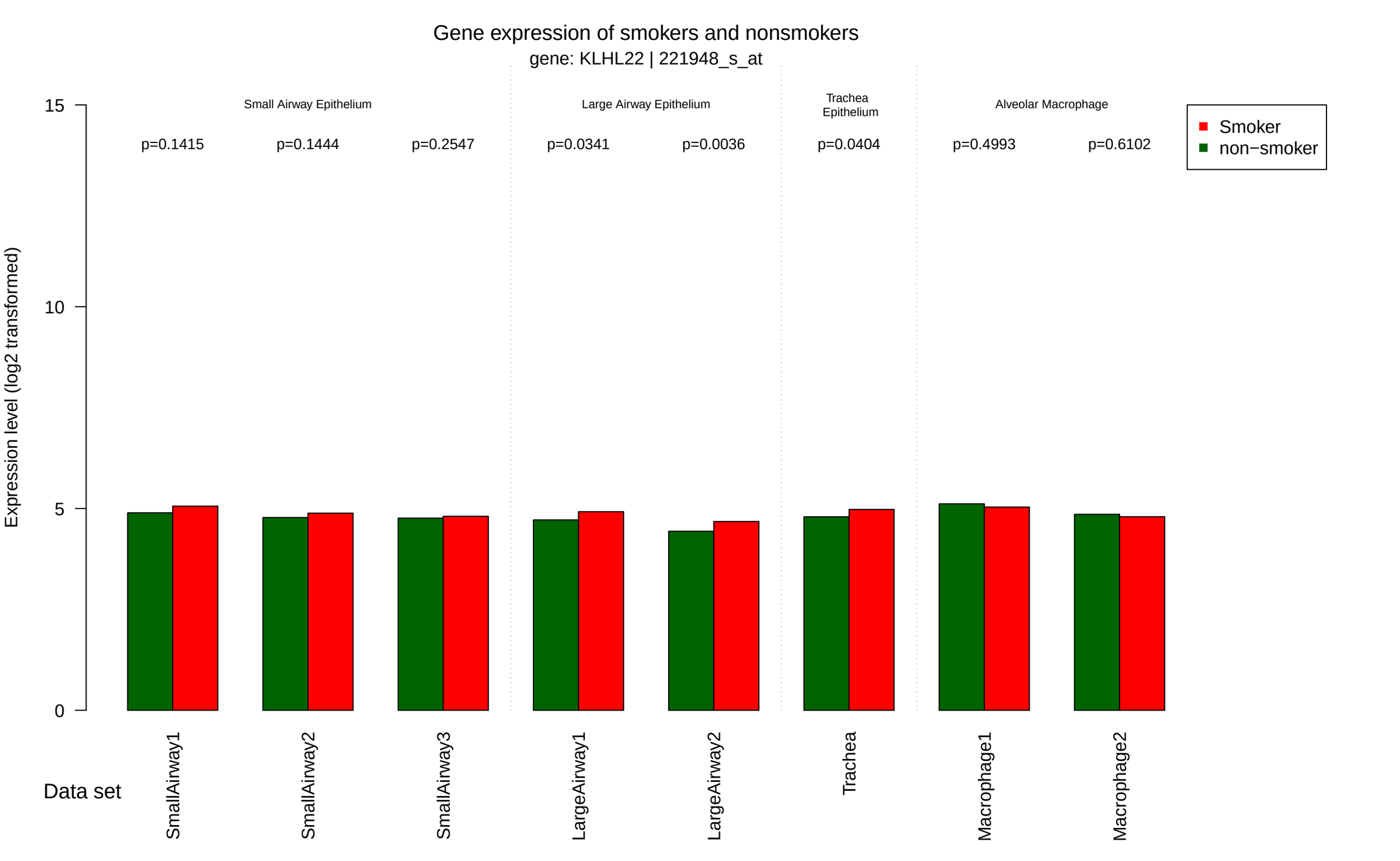
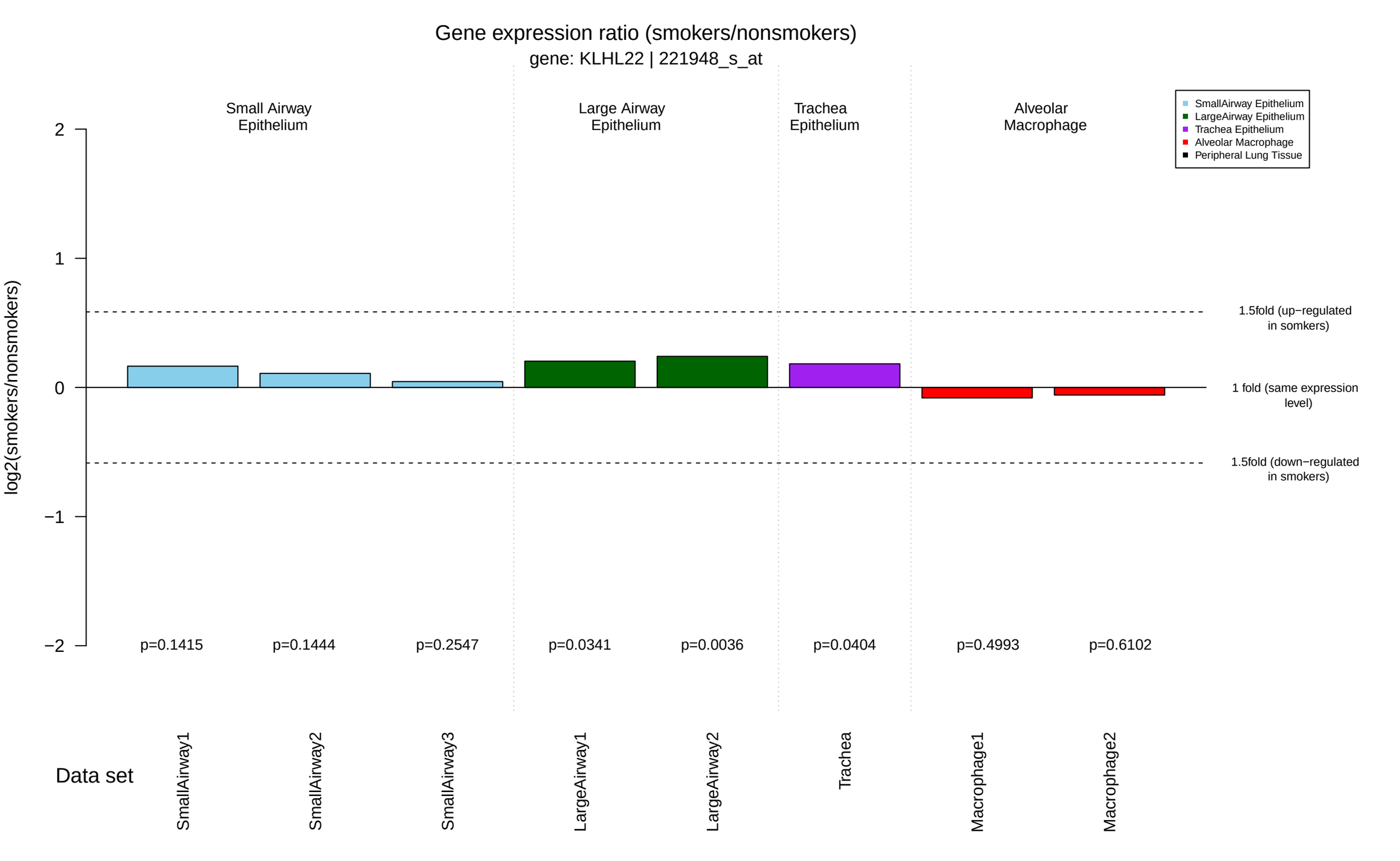
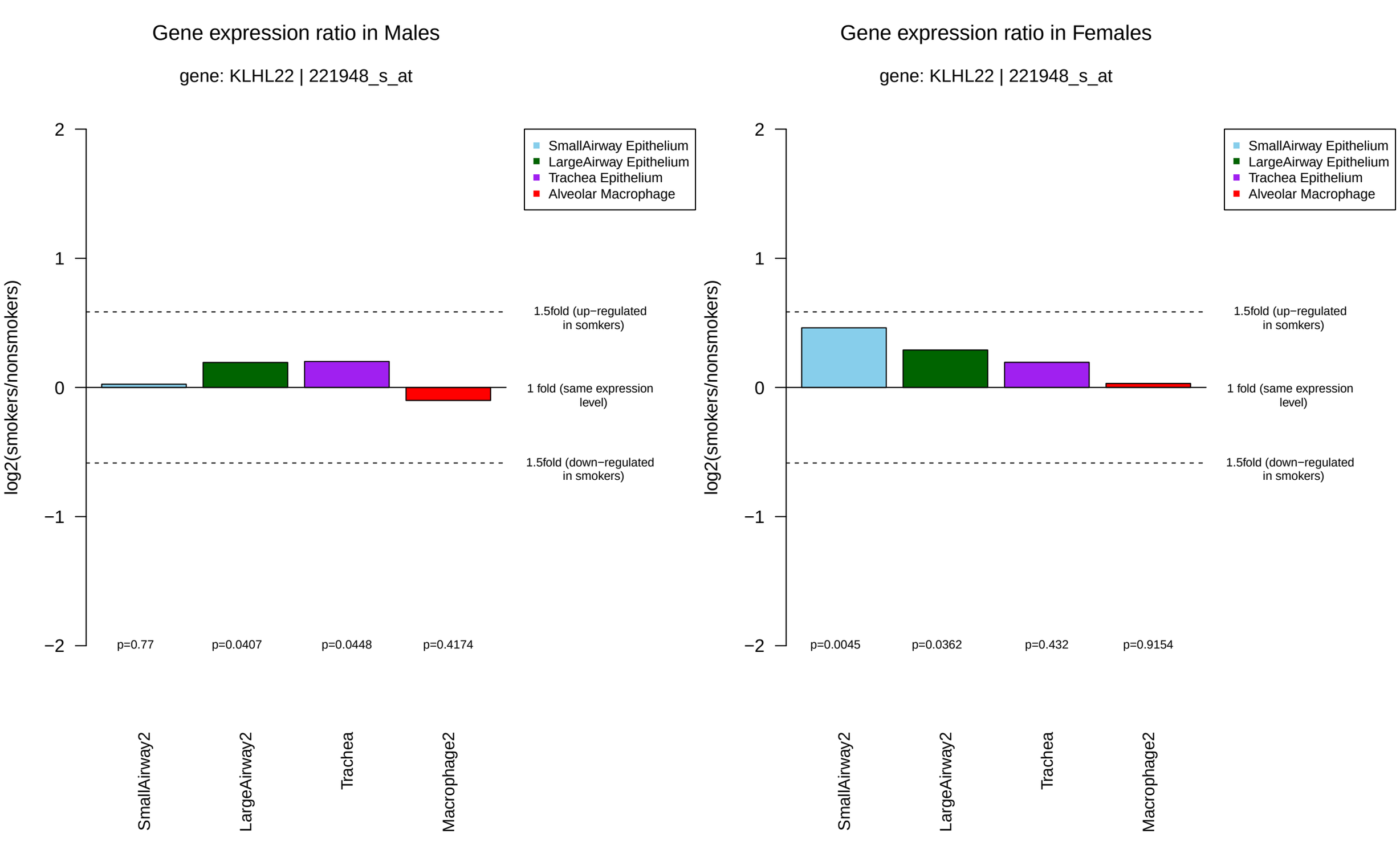
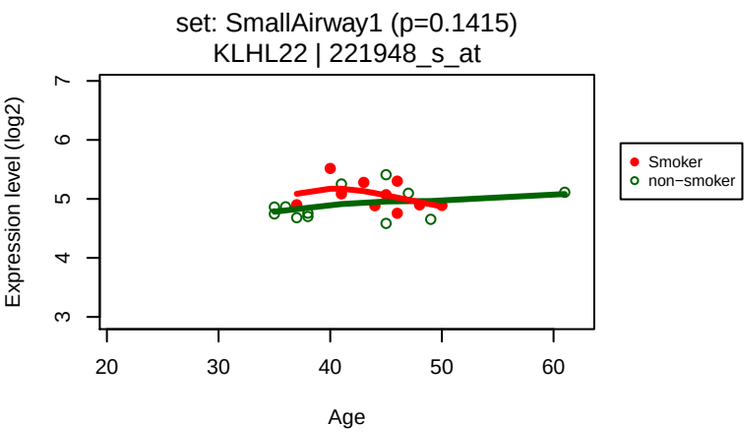
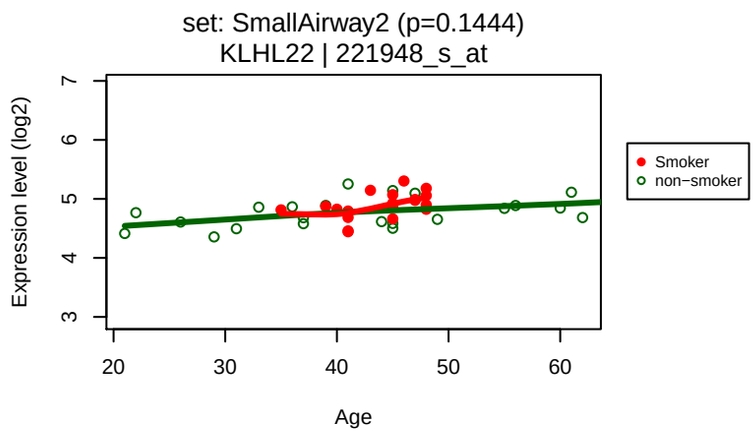
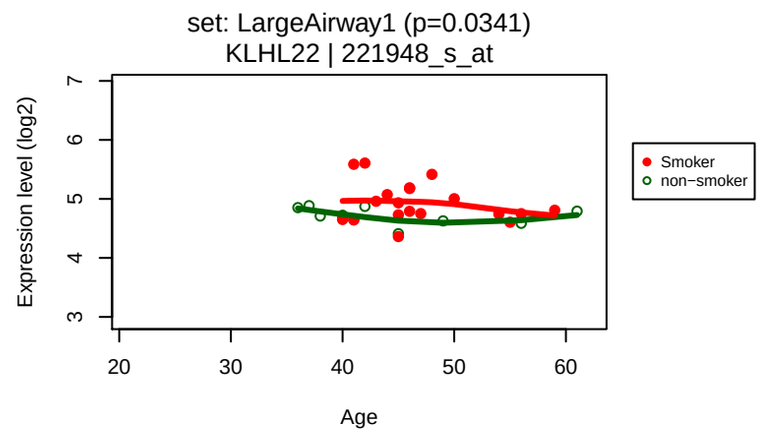

Gene expression of smokers and nonsmokers
gene: KLHL22 | 221948_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1415
p=0.1444
p=0.2547
p=0.0341
p=0.0036
p=0.0404
p=0.4993
p=0.6102
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

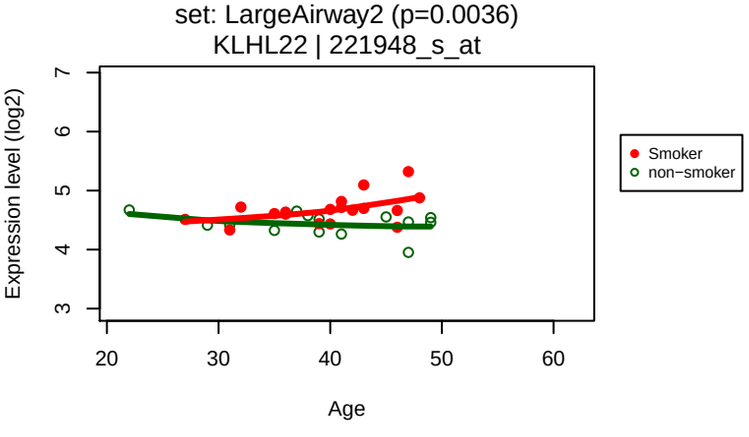

Gene expression ratio (smokers/nonsmokers)
gene: KLHL22 | 221948_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1415
p=0.1444
p=0.2547
p=0.0341
p=0.0036
p=0.0404
p=0.4993
p=0.6102
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

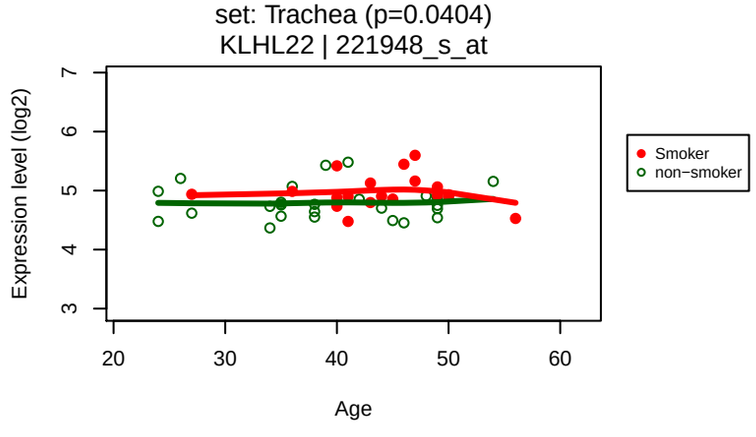

Gene expression ratio in Males
gene: KLHL22 | 221948_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.77
p=0.0407
p=0.0448
p=0.4174
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLHL22 | 221948_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0045
p=0.0362
p=0.432
p=0.9154
SmallAirway2
LargeAirway2
Trachea
Macrophage2

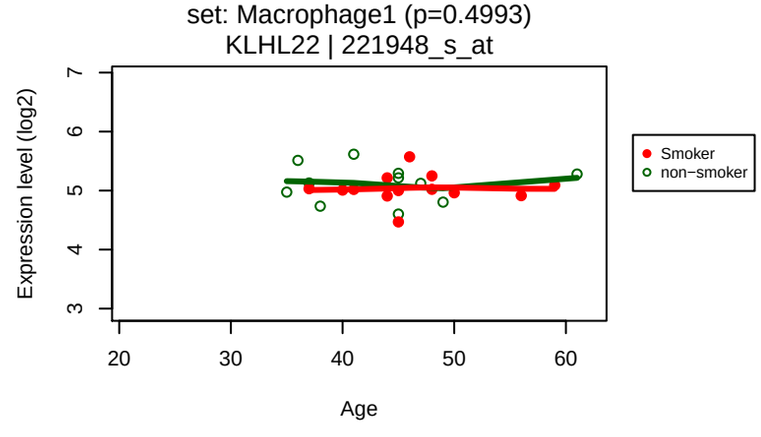

set: SmallAirway1 (p=0.1415)
KLHL22 | 221948_s_at
Expression level (log2)
Age
Smoker
non−smoker

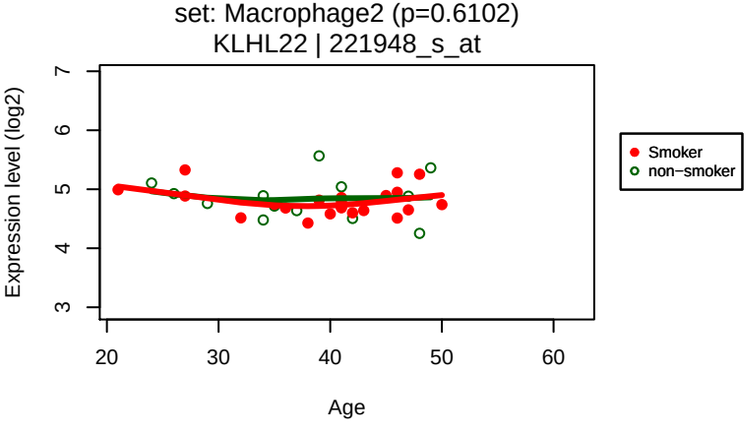

set: SmallAirway2 (p=0.1444)
KLHL22 | 221948_s_at
Expression level (log2)
Age
Smoker
non−smoker

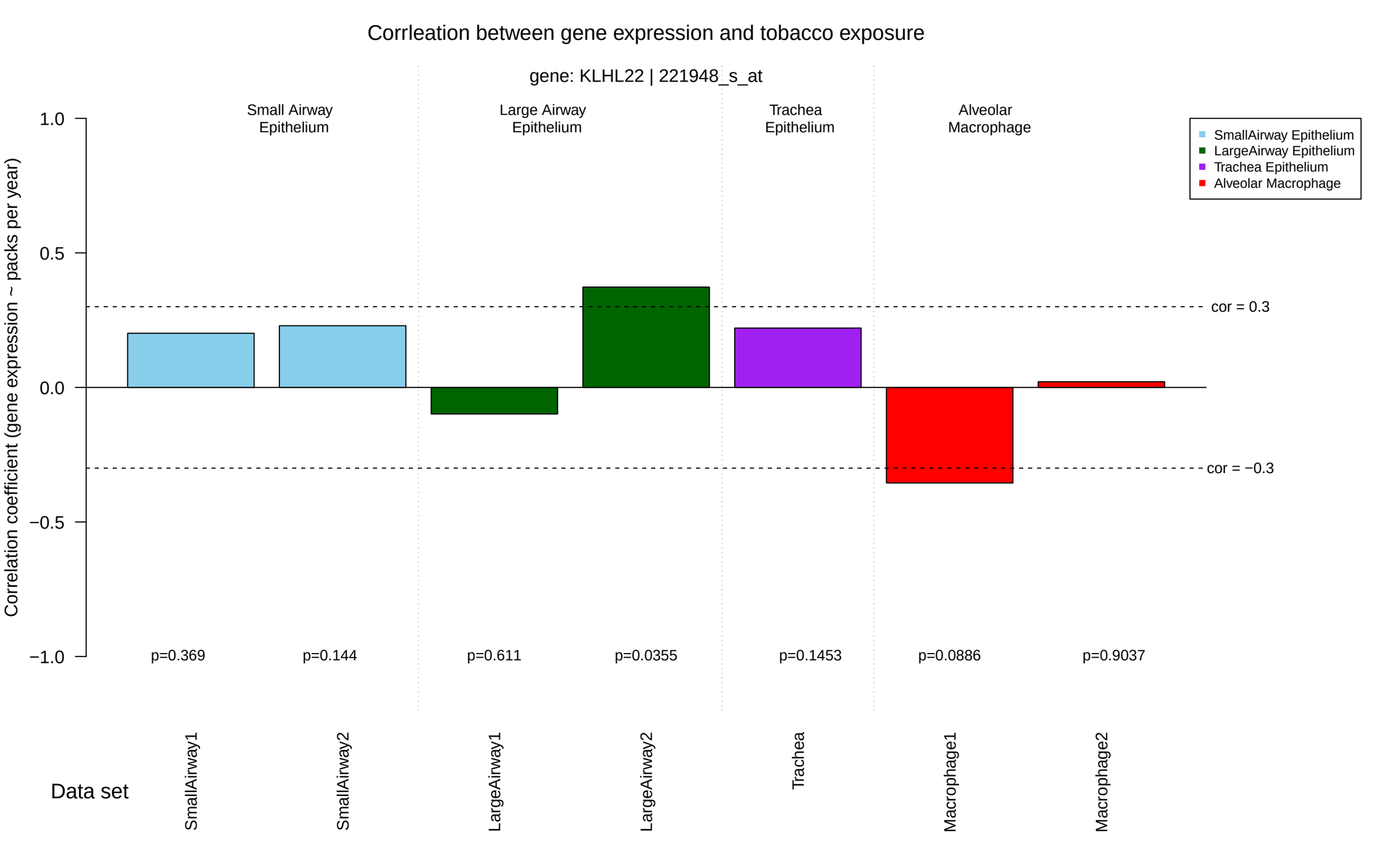

set: LargeAirway1 (p=0.0341)
KLHL22 | 221948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0036)
KLHL22 | 221948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0404)
KLHL22 | 221948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4993)
KLHL22 | 221948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6102)
KLHL22 | 221948_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLHL22 | 221948_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.369
p=0.144
p=0.611
p=0.0355
p=0.1453
p=0.0886
p=0.9037
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2